

PhantomFeel Dots: Augmenting the eating experience with thermal and vibrotactile stimuli on the face and throat

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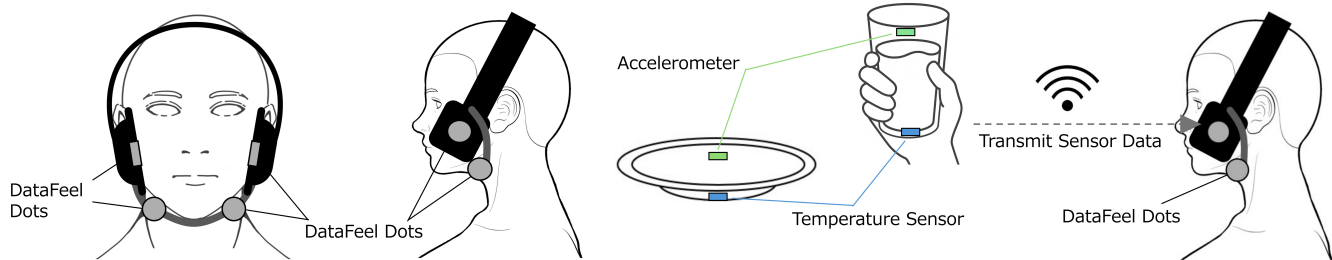


Figure 1: System Overview

Abstract

We propose PhantomFeel Dots, a wearable system that modulates sensation via thermal and vibrotactile stimuli applied to the cheeks and throat. Sensors added to the tableware detect its movement and temperature, and determine whether the user is eating or drinking. Based on this information, four DataFeel Dots worn on the user's cheeks and throat provide real-time thermal and haptic feedback. We expect that preheating the cheeks before eating enhances flavor perception, and that thermal and vibrotactile stimulation to the throat may contribute to a pleasant "throat feel." In addition, we plan to investigate whether gradually cooling the cheek dots and simultaneously heating the throat dots can evoke a thermal phantom sensation of warmth flowing from the mouth to the throat. This multisite, multimodal feedback enables dynamic modulation of eating experiences. We aim to offer rich, flexible eating experiences without modifying the food, tailored to individual sensory characteristics and dietary needs.

CCS Concepts

• **Human-centered computing** → *Human computer interaction (HCI)*; Haptic devices.

Keywords

multimodal feedback, thermal phantom sensation, haptics, eating experience, wearable interface

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1 Introduction

Eating is a major part of life, and improving the eating experience leads to a better quality of life. Recent HCI research has explored methods for enhancing eating experiences through sensory feedback [1, 3, 4, 6, 8, 10]. Most prior studies have focused on providing stimulation to a single site, such as the tongue [6, 8] or lips [3]. However, little attention has been given to the effectiveness of multi-site feedback or to how it can be designed to enrich the eating experience.

In response, we propose a system called PhantomFeel Dots that delivers coordinated thermal and vibrotactile stimuli via small wearable devices on the cheeks and throat, creating a sensation that flows from the cheeks to the throat. This system alters the perception of taste and temperature without modifying the actual food or drink, thereby offering a personalized extension of existing eating experiences. Therefore, it also holds potential to support individuals with dietary limitations due to health conditions, sensory hypersensitivity, or age-related declines in taste, serving as an alternative to traditional solutions such as sugar-free desserts or non-alcoholic beverages.

2 Related work

Various approaches have been proposed to use thermal and vibrotactile stimuli to alter the perception of taste and texture during eating.

In research using thermal stimulation, it has been demonstrated that taste sensations can be induced by applying stimuli to the

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tongue [4], nose [10], and lower lip [3]. In addition, a study [5] revealed that specific tastes, such as sweet and sour, alter blood flow in specific facial areas such as the cheeks and eyelids. This suggests that taste perception involves broader physiological activity across the face. Thus, stimulating the skin with thermal sources that affect blood flow may contribute to simulating or recreating aspects of these taste-related responses. Furthermore, a study involving thermal stimulation to the arm reported a phenomenon known as the "heat phantom sensation (h-PhS)" [9], in which applying thermal stimuli to two separate points induces the illusion of a heat source at the intermediate location. This suggests that thermal sensations can be perceived in regions other than the actual stimulation sites, indicating potential for applying such effects to enhance eating experiences.

Research also exists in the domain of haptics. A prior study [1] found a strong association between basic tastes (sweetness, sourness, umami, etc.) and specific vibrotactile frequencies, suggesting that vibrotactile stimulation can evoke or enhance taste imagery. Another study [7] showed that applying vibrotactile stimulation to the throat increases the frequency of swallowing. These findings suggest that vibrotactile stimulation affects the act of eating itself.

While prior studies suggest the potential of thermal or vibrotactile stimuli to influence taste, many have been limited to a single site, such as the tongue. We propose a novel approach to enhance taste by stimulating two sites on the cheeks and throat using DataFeel Dots [2]. The system induces a sensation of warmth moving from the mouth to the throat by coordinating the thermal stimuli applied to the cheeks and throat. Through this composite sensory presentation, we aim to enrich the eating experience.

3 PhantomFeel Dots

In this paper, we propose an interface design that realizes a richer eating experience by applying thermal and vibrotactile stimuli to four locations: both cheeks and the left and right sides of the throat. This section describes the components of the proposed system and provides an overview.

3.1 System Components

- **DataFeel Dots:** Used for detecting chewing and swallowing, and outputting thermal and vibrotactile stimuli.
- **Attachable Temperature and Accelerometer sensors for disposable plates and cups:** Used to detect the temperature of the food and when they are lifted.

3.2 System Overview

Figure 1 shows an overview of our proposed PhantomFeel Dots. PhantomFeel Dots is a wearable system that extends the eating experience—including taste, temperature sensation, and throat feel—by directly influencing the physical sensations of the person eating.

When the user picks up a plate or cup from the table, an accelerometer attached to it detects the movement and identifies which tableware was lifted, while a temperature sensor simultaneously measures the temperature of the food or drink. Based on this information, the system pre-heats or pre-cools the DataFeel Dots on both cheeks before the food reaches the mouth. This preliminary

stimulation enhances the anticipation of the flavor and temperature to be perceived in the mouth.

The IMU embedded in the cheek Dots automatically determines the user's state as "eating," "drinking," or "other." If the face tilts upward, it is classified as the beginning of drinking; otherwise, vertical motion of the cheeks is used to classify the action as the beginning of eating. Stimulus patterns optimized for each state are then generated. When the action is determined as "eating," the system monitors mastication through jaw movements detected by the IMU, continuing the thermal stimulation on the cheeks. Then, the thermal and vibrotactile stimuli are applied to the throat. On the other hand, when "drinking" is identified, a more rapid stimulus control is performed. Upon detecting a swallow, a stimulus is immediately applied to enhance the throat feel. Subsequently, the stimuli on the cheeks and throat are quickly attenuated to produce the characteristic crispness and refreshing sensation of a beverage.

We plan to enhance the sensation of an object passing from the mouth to the throat by modulating thermal presentation between the cheeks and the throat, with pre-defined temperature profiles. Figure 2 shows an example of these profiles. Previous studies [9] have explored a method known as "thermal phantom sensation (t-PhS)", in which a thermal illusion is created at an intermediate point between two heat sources. This phenomenon can induce a perceived movement of heat between two points. Therefore, it may be useful for expressing the sensation of heat sources moving between the cheeks and throat during eating and drinking. While t-PhS has been demonstrated on linear regions such as the arm, this study also attempts to evaluate the potential of t-PhS for simulating the sensation of objects passing through the throat.

This series of dynamic feedback mechanisms may also be applied to solve specific problems. For example, when drinking a non-alcoholic beverage, combining warming of the throat with rhythmic vibrotactile stimuli is expected to simulate the characteristic bite and stimulating, fizzy sensation of alcoholic beverages, providing high satisfaction. For hypersensitive individuals and older adults with reduced taste sensitivity, adjusting the thermal stimulus from the dots is expected to help them perceive taste more clearly.

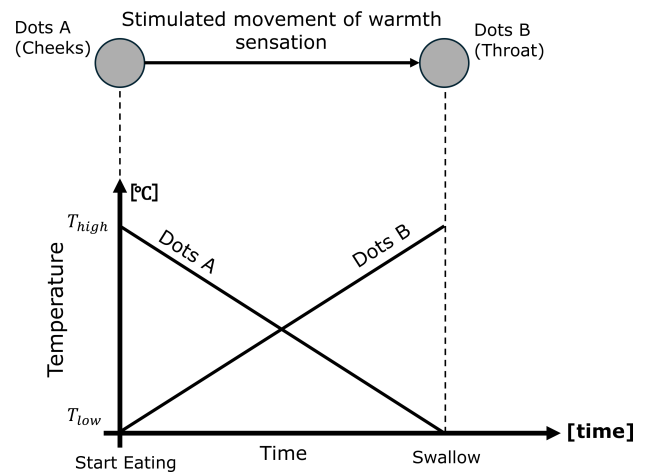


Figure 2: Temperature profile for enhancing warmth sensation during eating

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